A posteriori certification of a bounded pentagon word: Gaussian moments and validated overlap integrals

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Abstract

We continue reproducing a specified bounded Kim–Scarinci pentagon representation. Additional literature does not yet supply a directly matched kernel. A standard residual estimate reduces verification of saved finite Hermite chains to seven two-dimensional overlaps. Two chirp overlaps are evaluated by Gaussian moments in ball arithmetic. All five quantum-dilogarithm overlaps are certified using an existing Gauss–Legendre error theorem and Mehler bounds. Their combination gives $\|Df\| > 0.04489$ for the precisely defined $\hbar = 1$ continuous word and normalized Gaussian below. This is a computer-assisted statement about that literal spectral representation, not a finite-G observer correlator or a conclusion about the viability of de Sitter holography.

1 What the new literature does and does not supply

Belousov–Sarkissian–Spiridonov [2], equations (2.5), (4.2)–(4.5), (4.18), (5.9)–(5.11), and (6.16), retain an integer angular label. The binomial formula is a sum over that label and an integral over a continuous variable; its degeneration is taken as $b \to i$. Equation (4.12) requires real angular Fourier parameter and $\operatorname{Im} h > -1$. These are useful explicit formulae, but do not identify the fixed-deformation additive $\mathbb{R}^2$ phase below. In particular, the two real dimensions of a complex integration variable do not remove the discrete label of its Mellin–Fourier transform.

Andersen–Kashaev [3], equations (103), (117)–(122) and Lemma 6, use $A_N = \mathbb{R} \oplus \mathbb{Z}/N\mathbb{Z}$ with measure $N^{-1/2} \sum_n dx$, positive integer N, and $|\theta| = 1$, $\operatorname{Re} \theta > 0$. Their Fourier formula and finite Weyl operators require that cyclic spectral component. An explicit intertwining identification with the present representation is still missing. This is an applicability check, not a claim that no such identification can exist.

For validated integration we use existing Arb/Petras algorithms [4, 5], rather than inventing a new quadrature theorem. The caller must supply valid complex enclosures and check analyticity. Our calculation below does not claim a new general mathematical result.

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2 Residuals of actual saved vectors

Keep $\hbar = 1$, $P_j = -\pi i \partial_j$, $Q = (X, Y)$, $P = (P_y, P_x)$, and the phase $F = F_1$ defined in KS [1]. The operators

$$A = F(Q), \qquad B = F(P), \qquad M = e^{iXY/\pi}$$

are unitary on $L^2(\mathbb{R}^2, dx\,dy/(2\pi^2))$. The tested word is

$$D = AB - BM^{-1}BMA.$$

No heat multiplier is inserted in this section.

Let $v_0 = f$ and let v_j be any chosen finite vectors approximating $U_j v_{j-1}$. Define

$$e_j = \|U_j v_{j-1} - v_j\|.$$

Unitarity and the triangle inequality, iterated one step at a time, give

$$\|U_m \cdots U_1 f - v_m\| \le \sum_{j=1}^{m} e_j.$$

For saved left and right chains $l_0 = r_0 = f$ this yields

$$\|Df\| \ge \|l_2 - r_5\| - \sum_{j=1}^{2} e_{L,j} - \sum_{j=1}^{5} e_{R,j}. \tag{1}$$

Every step error is determined by one overlap:

$$e_j^2 = \|v_{j-1}\|^2 + \|v_j\|^2 - 2\,\mathrm{Re}\langle v_j, U_j v_{j-1}\rangle. \tag{2}$$

These statements do not require the saved vectors to be exact orthogonal projections. Consequently they avoid interpreting a quadrature-dependent norm loss as an already certified leakage.

The left steps are (B, A); the right steps are (A, M, B, M^{-1}, B). For a B step, Fourier transform both finite vectors. The Hermite Fourier factors are exactly $(-i)^{m+n}$, so the remaining overlap is again a coordinate integral, with multiplier $F(y, x)$. At $\hbar = 1$ the exact phase is symmetric. Thus five F overlaps and two chirp overlaps suffice, without nested Fourier kernels or an assumption that a single chiral factor acts on the Gaussian.

All saved complex binary64 coefficients are lifted as exact dyadic numbers. They specify legitimate finite vectors independently of how accurately they approximated a projection. The initial state is the exact normalized oscillator vacuum, not a subsequently rounded normalization.

3 Gaussian moments certify the chirp steps

In oscillator coordinates $z = x/\sqrt{\pi}$, $t = y/\sqrt{\pi}$, a normalized Hermite expansion has the form

$$v(z,t) = \pi^{-1/2} e^{-(z^2+t^2)/2} \sum_{m,n} c_{mn} p_m(z) p_n(t), \qquad p_m(z) = \frac{H_m(z)}{\sqrt{2^m m!}}.$$

The coefficient norm is its Hilbert norm. Multiplication by M^s, $s = \pm 1$, inserts e^{istz}. For polynomials V, W specified by these coefficients,

$$\langle w, M^s v \rangle = \frac{1}{\pi} \int_{\mathbb{R}^2} e^{-z^2-t^2+istz}\, \overline{W(z,t)} V(z,t)\, dz\, dt.$$

Conjugation here acts on polynomial coefficients on the real plane. The Gaussian prefactor is $2/\sqrt{5}$ and the normalized covariance is

$$C_s = \begin{pmatrix} 2/5 & is/5 \\ is/5 & 2/5 \end{pmatrix}.$$

Writing its normalized moments as μ_{mn}, one has

$$\mu_{00} = 1, \quad \mu_{0,2k} = (2k-1)!!(2/5)^k, \quad \mu_{0,2k+1} = 0,$$

$$\mu_{mn} = (m-1)\frac{2}{5}\mu_{m-2,n} + n\frac{is}{5}\mu_{m-1,n-1} \quad (m \geq 1).$$

A finite polynomial product therefore gives the overlap without quadrature, boundary truncation or spectral-tail approximation. Ball arithmetic bounds rounding and cancellation in this evaluation.

Hermite size	saved quadrature	$e_{R,2}$	$e_{R,4}$
64	512	0.00400544475824	0.01177031292439
128	1024	0.00234776595683	0.00787506108862

These displayed decimals abbreviate stored rational endpoints. The first case uses 512-bit arithmetic; the second was evaluated at 768 and 1024 bits. Independent integer Wick pairing sums check selected moments through degree $(254, 254)$. The vacuum control also uses a one-dimensional Arb Gaussian integral. An initially missing factor two in that control's two-sided tail bound was corrected; the initial output is retained, and the chirp overlaps are unchanged.

4 A quantitative conditional target

Suppose, for each of the five full F overlaps,

$$\mathrm{Re}\langle v_j, U_j v_{j-1} \rangle \geq \|v_j\|^2 - \eta.$$

Then (2) gives $e_j \leq \sqrt{\|v_{j-1}\|^2 - \|v_j\|^2 + 2\eta}$ for these particular saved vectors. For the size-128 chain, (1) would give respectively 0.05208026006 at $\eta = 10^{-5}$ and 0.02048259469 at $\eta = 10^{-4}$. These were initially conditional targets. The completed calculation below proves all five hypotheses for $\eta = 10^{-4}$, but not for $\eta = 10^{-5}$, and gives a stronger bound using their actual individual error enclosures.

5 A validated local pilot and the full-plane task

For $\hbar = 1$ use the finite product

$$F_N(x,y) = \prod_{j=0}^{N-1} \frac{1+e^{x-(2j+1)\pi-iy}}{1+e^{x-(2j+1)\pi+iy}} \frac{1+e^{y-(2j+1)\pi-ix}}{1+e^{y-(2j+1)\pi+ix}}.$$

For real $|x|, |y| \le L_x$ with $r = e^{L_x-(2N+1)\pi} < 1$, its exact-arithmetic tail obeys

$$|F - F_N| \le \frac{4r}{(1-e^{-2\pi})(1-r)}.$$

On a rectangle avoiding the product singularities, the remaining finite integrand is meromorphic in its complexified variables. In a nested integral with real inner coordinate z, poles require $\operatorname{Re} t$ to lie on an odd multiple of $\sqrt{\pi}$. The outer callback therefore rejects any complex ball intersecting one of these rows before asserting analyticity. It never applies ordinary conjugation or absolute value to a complex integration variable.

A pilot for the size-64 first right step on $(z,t) \in [-1,1]^2$, $N = 8$, encloses the finite-product overlap near

$$0.710145049 - 0.0000006487\, i.$$

The stored enclosure has sub-10^{-9} width; the product tail is separately bounded. This is one compact square, not the full overlap.

For the size-128 first step, the input is exactly Gaussian. Outside $[-L, L]^2$, Cauchy–Schwarz bounds the omitted overlap by

$$\sqrt{2\operatorname{erfc}(L)}\,\|w\|.$$

Removing horizontal strips of half-width ρ around K pole rows costs at most $2K\rho\,\mathcal{K}_n\|w\|$, where the standard Mehler formula gives, for $0 < r_0 < 1$,

$$\mathcal{K}_n = \frac{r_0^{-(n-1)}}{\sqrt{\pi(1-r_0^2)}}.$$

This follows by bounding the one-coordinate Hermite projection kernel and applying Cauchy–Schwarz in the other coordinate. Together with the product tail it gives an explicit omission budget. This nested approach was discontinued; the tensor method below certifies the first full overlap. Remaining steps need their own tail bounds and complete overlap enclosures.

6 A uniform outer bound for nested integration

There is a computational distinction between evaluating the inner integral at a nearly exact outer node and bounding it on an entire complex ball. Asking the inner integrator to resolve parameter uncertainty to a tiny tolerance can waste subdivisions. For the Gaussian-input step, Mehler's formula [6] supplies the following uniform bound instead.

Let $I_N(t)$ be the inner z integral of the finite-product Gaussian overlap, over any finite real interval. Set $u = \sqrt{\pi}\,\mathrm{Re}\,t$, $v = \sqrt{\pi}\,\mathrm{Im}\,t$, $a_j = (2j+1)\pi$, and

$$G_N(u,v) = \frac{e^{2N|v|}}{|\cos(u/2)|^N} \prod_{j=0}^{N-1} \frac{1+e^{u-a_j}}{|1-e^{u-a_j}|}.$$

This is finite away from the excluded pole rows. For real x, the first product ratio is bounded by $e^{2|v|}/|\cos(u/2)|$, since $|1+Re^{iu}| \geq (1+R)|\cos(u/2)|$ for $R \geq 0$. The second ratio is at most $(1+e^{u-a_j})/|1-e^{u-a_j}|$ by the triangle inequalities. These estimates retain all finite factors.

For any $0 < r < 1$, the finite Hermite polynomial kernel satisfies

$$\sum_{k=0}^{n-1} |p_k(z)|^2 \leq \frac{r^{-(n-1)}}{\sqrt{1-r^2}} \exp\left(\frac{2r(\mathrm{Re}\,z)^2}{1+r} + \frac{2r(\mathrm{Im}\,z)^2}{1-r}\right).$$

Apply this estimate to both coordinates, use coefficient Cauchy–Schwarz for W, multiply by the vacuum Gaussian, and extend the absolute inner Gaussian integral to the real line. The result is

$$|I_N(t)| \leq \frac{\|w\| r^{-(n-1)}}{\sqrt{\pi(1-r)}} \exp\left(-\frac{(\mathrm{Re}\,t)^2}{1+r} + \frac{(\mathrm{Im}\,t)^2}{1-r}\right) G_N(u,v).$$

Each of several chosen r values gives a valid bound, so the smallest upper endpoint may be used. The outer callback returns a containing complex ball from this inequality for its analytic-bound requests; nearly exact quadrature nodes still use actual inner integration. Holomorphy is checked by the pole-row guard, not inferred from a finite numerical answer. This nested run was subsequently stopped without a completed segment; the direct tensor method below replaces it.

7 Direct use of an existing quadrature theorem

Johansson's Section 2 [4] states that an analytic function bounded by M on a Bernstein ellipse of parameter $\rho > 1.1$ has n_q-point Gauss–Legendre error at most $50M\rho^{-2n_q}$. We use this existing result with $\rho = 2$, whose semiaxes are $5/4, 3/4$; we do not claim a new quadrature theorem.

For a rectangle with half-widths h_x, h_y, let M_x bound the integrand on the x ellipse times the real y interval, and let M_y bound it on the real x interval times the y ellipse. Positivity and sum 2 of the reference weights, together with

$$I_x I_y - Q_x Q_y = (I_x - Q_x)I_y + Q_x(I_y - Q_y),$$

give the tensor error bound

$$E_R = 100 h_x h_y \, 2^{-2n_q}(M_x + M_y).$$

The scaled integrand is the analytic extension of $\overline{w(z,t)}F_N(\sqrt{\pi}z, \sqrt{\pi}t)h_0(z)h_0(t)$: only coefficients are conjugated in its extension. Every product denominator must exclude zero on the bounding complex rectangle. Otherwise no analytic quadrature bound is accepted.

For a saved $n \times n$ vector, Mehler's formula and coefficient Cauchy–Schwarz give the following amplitude bound for any $0 < r < 1$:

$$|\overline{w}_{\mathrm{an}}(z,t)h_0(z)h_0(t)| \le \frac{\|w\| r^{-(n-1)}}{\pi\sqrt{1-r^2}} \exp\left[-\frac{(\Re z)^2 + (\Re t)^2}{1+r} + \frac{(\Im z)^2 + (\Im t)^2}{1-r}\right].$$

The code chooses among eight valid values $r = 1 - 2^{-k}$. On real cells $|F_N| = 1$ almost everywhere. A cell touching a singularity can therefore contribute zero to the numerical sum while its full absolute integral bound is added to the error. No region is silently deleted. Dyadic subdivision covers the whole square exactly.

For $[-L, L]^2$, the remaining errors are bounded by

$$T \le \|w\|\sqrt{2\,\mathrm{erfc}(L)}, \qquad P \le \frac{4 r_N \|w\|}{(1 - e^{-2\pi})(1 - r_N)}, \quad r_N = e^{L\sqrt{\pi} - (2N+1)\pi} < 1.$$

The first bound uses the exact vacuum input and Cauchy–Schwarz; it must not be transferred to a general finite Hermite input.

8 The first certified overlap and its checks

The first right step, $\langle w, Af\rangle$, used the saved $n = 128$ vector, $L = 5$, $N = 3$, $n_q = 24$ and 256-bit ball arithmetic. There were 925 cells, including 27 absolute-bound cells. The bounds are

$$E < 1.940\,10^{-5}, \quad T < 1.754\,10^{-6}, \quad P < 7.963\,10^{-6}.$$

Including ball rounding, the saved output proves

$$\boxed{\|w\|^2 - \Re\langle w, Af\rangle < 2.973\,10^{-5} < 10^{-4}.}$$

The run completed normally in about 26 seconds. This first calculation alone does not decide the word; the complete chain is reported below.

Three independent low-Hermite antiderivative controls include complex coefficients and asymmetric rectangles. Two further comparisons agree with the earlier nested-Arb rectangle. The initial test incorrectly required a narrower enclosure to contain a wider one; the corrected criterion is intersection, and both original failures are retained. Exact-rational coverage and error accounting, all 925 bound replays, and selected 320-bit quadrature checks pass 2794 checks. An initial audit incorrectly compared precision-dependent upper endpoints as enclosures of one common exact number. Replaying bounds at their original precision corrects that check; higher-precision integration comparisons remain separate. These tests are not formal verification of the arithmetic library.

9 General finite pairs: weighted Mehler bounds

For a finite coefficient array c define, for $0 < r < 1$,

$$A_c(r) = \sum_{m,n} |c_{mn}|^2 r^{-m-n}, \quad X_c(r) = \sum_{m,n} |c_{mn}|^2 r^{-m}, \quad Y_c(r) = \sum_{m,n} |c_{mn}|^2 r^{-n}, \quad a_r = \frac{1-r}{1+r}.$$

Weighted Cauchy–Schwarz and the standard Mehler formula give

$$|\overline{w}_{\mathrm{an}}(z,t)v(z,t)| \le \frac{\sqrt{A_v(r)A_w(r)}}{\pi(1-r^2)} \exp\left[-a_r((\Re z)^2 + (\Re t)^2) + a_r^{-1}((\Im z)^2 + (\Im t)^2)\right].$$

For the exterior of a real square, integrate the other coordinate first using Hermite orthogonality, and apply the one-coordinate Mehler bound. The union bound gives

$$T_c(L) := \|1_{\mathbb{R}^2 \setminus [-L,L]^2} c\|^2 \le \frac{X_c(r) + Y_c(r)}{1-r} \operatorname{erfc}(\sqrt{a_r} L).$$

Each chosen r is a valid bound. Minimize the upper endpoints separately for v, w; the omitted overlap is at most $\sqrt{T_v(L)T_w(L)}$. This handles the entire finite vectors, including every retained high Hermite mode, without treating them as Gaussian inputs. It is an application of existing identities, not a new general theorem.

A sharper phase enclosure. For real u, b, y write $x = u + ib$, $a_j = (2j+1)\pi$, and

$$d(s,\theta) = |1 + e^{s+i\theta}|e^{-s} = \sqrt{(e^{-s}-1)^2 + 4e^{-s}\cos^2(\theta/2)}.$$

Subtract the conjugate denominator from the numerator of each factor. The triangle inequality then yields

$$|F_N(u+ib, y)| \le \prod_{j=0}^{N-1} \left(1 + \frac{2|\sin b|}{d(u-a_j, b+y)}\right)\left(1 + \frac{|e^{2b}-1|}{d(y-a_j-b, u)}\right).$$

For a y ellipse interchange coordinates, using the symmetry of F_N at $\hbar = 1$. All denominator lower bounds must be strictly positive. Zero-containing bounds reject analytic quadrature and trigger subdivision or a real absolute-integral bound. The initial implementation used interval powers across zero, which can return NaN in this interface. The production version squares nonnegative lower endpoints, preserving rigorous denominator bounds.

Controls. Eleven checks cover three non-vacuum complex-coefficient integrals against explicit low-Hermite antiderivatives, two exact Fourier norms, four complex-point phase inequalities, rejection at a pole, and a non-vacuum exterior mass against incomplete Gaussian moments. These controls supplement the analytic derivation; they do not replace it or independently verify the numerical library.

10 Completed chain and a positive continuous bound

All five overlaps have now completed normally. The three non-vacuum steps use $L = 16$, eight product factors, 24-point tensor quadrature, and 256-bit ball arithmetic. Their spatial overlap tails are less than $1.315\,10^{-6}$, $1.052\,10^{-6}$, and $3.000\,10^{-6}$, respectively. The table gives safe upward decimal bounds, not fitted errors. The "deficit" means $\|w\|^2 - \Re\langle w, Uv\rangle$.

Step	cells	deficit upper bound	e_j upper bound
$L, 1$	925	$2.973\,10^{-5}$	0.010410
$L, 2$	6229	$2.364\,10^{-5}$	0.015491
$R, 1$	925	$2.973\,10^{-5}$	0.010410
$R, 3$	6157	$2.322\,10^{-5}$	0.015776
$R, 5$	6907	$2.417\,10^{-5}$	0.012427

The total of 21,143 cells includes absolute-bound cells covering singular regions. All omitted mass, product tails, quadrature errors, and ball rounding enter the reported bounds. The first left and first right steps are related by Fourier symmetry; they are distinct chain errors, not independent physical measurements.

Using the stored rational endpoints rather than the rounded table, the exact finite-vector separation and the upper bound on all seven step errors satisfy

$$\|l_2 - r_5\| > 0.1196309251, \qquad \sum e_j < 0.0747344165.$$

Consequently the a posteriori inequality proves

$$\boxed{\|(AB - BM^{-1}BMA)f\| > 0.04489 > 0.}$$

This is a positive bound for the continuous operators. It does not assert that their exact residual equals this lower bound.

An exact-rational audit checks cell coverage, disjointness, all recorded sums and the worst-endpoint inequality. It also replays every analytic/direct cell bound at its original precision and selected quadratures at 320 bits: 63,551 checks pass. These checks share the stated mathematics and library, and are not 63,551 independent proofs. A further actual size-128 right-step-5 rectangle was computed using polynomial evaluation and nested Arb integration; its real and imaginary enclosures agree with the tensor method. The earlier failed development checks are retained with their fixes.

11 The precise continuous representation being tested

The state in the original measure is

$$f(x,y) = \sqrt{2}\, e^{-(x^2+y^2)/(2\pi)}, \qquad \int_{\mathbb{R}^2} |f|^2 \frac{dx\, dy}{2\pi^2} = 1.$$

The unitary coordinate change to the oscillator convention sends this state to $h_0(z)h_0(t)$, with $z = x/\sqrt{\pi}, t = y/\sqrt{\pi}$. The infinite product is the almost-everywhere limit of the displayed F_N. Values on its discrete singular set can be assigned arbitrarily with modulus one without changing the multiplication operator.

The source conventions were checked again against KS Section 6.3, equation (35), and Proposition 6.15, and against the pure-imaginary parameter formulas in Kim's survey, Section 3 [7]. With $q = e^{-\pi}$ the two compact-product ratios give precisely the displayed symmetric $F(x,y)$ at $\hbar = 1$; this check does not use a formal cancellation of unbounded operators.

For $P_x = -\pi i \partial_x$, the exact translation action gives

$$e^{isX} e^{itP_x} = e^{-i\pi st} e^{itP_x} e^{isX}.$$

The analogous formula holds for (Y, P_y); the other coordinate and momentum pairs commute strongly. Consequently the ordered quadruple (X, Y, P_y, P_x) obeys the Weyl assumptions stated in Proposition 6.15. Moreover, by direct differentiation and unitary conjugation,

$$M^{-1} P_y M = P_y + X, \qquad M^{-1} P_x M = P_x + Y.$$

The two sums strongly commute since they are simultaneous unitary conjugates of P_y, P_x. Bounded joint spectral calculus therefore identifies $M^{-1}BM$ with $F(X + P_y, Y + P_x)$.

These identities specify the literal continuous test independently of any finite matrix commutators. The finite vectors are only auxiliary comparison vectors in the residual inequality. If their certified residual sum is below their final separation, the conclusion concerns this exact bounded word on $L^2(\mathbb{R}^2)$. Its comparison with the published assertion must retain the explicit definitions above and the distinction between a literal spectral operator and a proposed analytic continuation prescription.

No observable time, observer mass, gravitational coupling G, or physical four-point gluing measure has been assigned to these operators. A nonzero word would obstruct using this literal implementation as an already verified physical building block; it would not rule out de Sitter holography or every other representation.

References

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[6] NIST DLMF, equation 18.18.28 (Mehler formula).

[7] H. K. Kim, arXiv:2308.02934v2, Section 3.